

Cognitive-Behavioral Treatment of Generalized Anxiety Disorder Among Adolescents: A Case Series

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ABSTRACT

Seven adolescents received a cognitive-behavioral treatment targeting generalized anxiety disorder. The treatment consisted of awareness training, worry interventions, and relapse prevention. The worry interventions targeted specifically intolerance of uncertainty, beliefs about worry, problem solving, and cognitive avoidance. According to the Anxiety Disorders Interview Schedule for *DSM-IV* diagnosis of generalized anxiety disorder, self-report questionnaire scores, and time spent worrying every day, three adolescents showed clinically significant change at post-test which was maintained at 6- and 12-month follow-up assessments. Minimal to moderate improvement was observed for other participants. Factors that may explain these differences are discussed. *J. Am. Acad. Child Adolesc. Psychiatry*, 2003, 42(3):327–330.

Key Words: worry, generalized anxiety disorder, adolescents.

Recent epidemiological studies show that generalized anxiety disorder (GAD) is one of the most prevalent anxiety disorders (Kessler et al., 1994). With the advent of the *DSM-IV* (American Psychiatric Association, 1994), the diagnosis of GAD replaced the overanxious disorder of childhood, thus indicating a clear continuity between the childhood disorder and adult GAD. Studies exploring the age at onset of GAD have found that adult patients often spontaneously report having been worriers all their lives. In most patients, GAD develops gradually during adolescence and reaches clinical significance during young adulthood (Léger et al., 1999; Rapee, 1991; Yonkers et al., 1996). Even though GAD, together with specific phobias, is the anxiety disorder that occurs most frequently before the age of 18 (Scheibe and Albus, 1992), very few studies have examined the characteristics of GAD among children or adolescents.

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Based on a model of excessive worry conceptualized by Dugas and colleagues (1998), Laugesen and Dugas (2000) studied the presence of four worry-related cognitive processes in an adolescent sample. Their results showed that intolerance of uncertainty, positive beliefs about worry, negative problem orientation, and cognitive avoidance were all significantly correlated with the tendency to worry among high school adolescents. Furthermore, intolerance of uncertainty, positive beliefs about worry, and negative problem orientation each made a unique and significant contribution to the prediction of worry. The results of this study suggest that this model of GAD originally developed to explain adult worry can also be applied to adolescents.

The study of the efficacy of psychosocial treatments for anxiety among children and adolescents is still a new area. To date, two randomized clinical trials have used cognitive-behavioral interventions to target some anxiety disorders in children and adolescents (Kendall, 1994; Silverman et al., 1999). The aim of the present study was thus to explore the possible application to adolescents of a cognitive-behavioral treatment for GAD (see Ladouceur et al., 2000).

METHOD

Participants

Seven adolescents who responded to an advertisement in a local newspaper were included in this study. They were four girls (aged 17,

17, 17, and 18 years) and three boys (aged 16, 18, and 18 years), and all were French Canadian. The principal entry criteria were (1) age 14 to 18 and (2) primary diagnosis of GAD. All participants, and parents for participants under 18, gave consent for this report to be published.

Measures

Diagnostic Assessment. The Anxiety Disorders Interview Schedule for *DSM-IV* (ADIS-IV) (Brown et al., 1994) was used to diagnose participants and evaluate treatment outcome. The pretreatment ADIS-IV was conducted by the therapist. An independent assessor who was not involved in pretreatment assessment or in treatment conducted posttreatment and follow-up ADIS-IV interviews; this assessor was the same for all participants. The assessors rated the severity of the diagnosis on a 9-point scale ranging from 0 to 8. Conventionally, clinician severity ratings of 4 through 8 represent diagnoses above the clinical threshold.

Questionnaires. The Penn State Worry Questionnaire (Meyer et al., 1990; translated by Ladouceur et al., 1992) consists of 16 items that measure a trait-like tendency to worry. The Worry and Anxiety Questionnaire (Dugas et al., 2001) is a 10-item questionnaire that evaluates *DSM-IV* diagnostic criteria for GAD. Only the somatic symptoms section (six items) is reported here. The Worry Domains Questionnaire-Adolescent form (WDQ) (Tallis et al., 1992; adapted for adolescents by Fournier et al., 1995) contains the original 30 items and 15 additional items related to teenagers' experience. Examples of the covered themes are school, dates, and alcohol or drug consumption. Intensity of worry for each item is rated on a Likert-type scale ranging from 0 (not at all) to 4 (extremely). The WDQ yields a global worry score that includes both the range and severity of worry. The Beck Anxiety Inventory (Beck et al., 1988; translated by Freeston et al., 1994) is a 21-item state anxiety scale that measures the intensity of cognitive, affective, and somatic anxiety symptoms experienced over the past 7 days. The Children's Depression Inventory (Kovacs, 1983; translated by Saint-Laurent, 1990) is an age-adapted version of the Beck Depression Inventory, consisting of 27 items covering the principal symptoms of depression.

Daily Self-Monitoring. Daily self-monitoring was done during baseline, treatment, and follow-up assessments. Time spent worrying each day was compared to a personal maximum of worry for a single day that had been established with the therapist at the beginning of the baseline. The resulting proportion was recorded in a booklet at the end of each day.

Treatment

A single therapist treated all participants. One-hour therapy sessions were conducted weekly. The average treatment length for the six treatment completers was 13.2 sessions. The treatment consisted of awareness training, worry interventions, and relapse prevention. The worry interventions targeted the four cognitive components of the Dugas et al. (1998) model of GAD. First, participants identified their own manifestations of intolerance of uncertainty in their everyday lives. They then chose simple behaviors that they performed to increase their tolerance of uncertainty (e.g., ask a teacher a question to get information about a class subject, even if I am not sure what the teacher's reaction will be). Second, cognitive techniques were used to help participants reevaluate their beliefs about worry, which primarily involved the perceived usefulness of worrying. Third, problem solving training was used to address worries about current problems. The therapist taught participants the steps in the problem-solving process, with an emphasis on the first step, problem orientation. It was explained that poor problem orientation often takes the form of seeing a problem as

a threat, which impedes the resolution of the problem. Participants were invited to look at a problem as if it were a challenge by asking questions such as "What can I learn, get, or even gain from this situation?" Fourth, cognitive exposure targeted the avoidance of the threatening mental images often related to key worry topics about highly remote events. Participants described a worrisome scenario, which was recorded for exposure. Repeated exposure was then applied with a Walkman tape recorder while using covert response prevention.

RESULTS AND DISCUSSION

Table 1 presents scores on the different measures at each assessment time. The clinical significance of change was based on the evaluation of treatment response, following the lines of Ladouceur et al. (2000). For self-report questionnaires and daily self-monitoring of worry, treatment response was defined by a 20% change in pretreatment scores. For the GAD residual score on the ADIS-IV, a score of 3 or less was considered to reflect treatment response. For each participant, responder status was then determined as follows: criteria reached on 0 to 1 measures = low, on 2 to 4 measures = moderate, and on 5 to 7 measures = high. Table 2 presents the number of participants in each category of responder status at each assessment time.

Based on the ADIS-IV evaluation of GAD diagnostic criteria, self-report questionnaires, and daily self-monitoring of worry, three participants (participants 1, 4, and 7) showed a large decrease in GAD symptoms which was maintained at 6- and 12-month follow-up assessments. Participants 2 and 5 showed a moderate decrease in GAD symptoms, and the improvement attained was partly maintained at both follow-up assessments. Participant 3 stayed unchanged at post-test and failed to complete the follow-up assessments, and participant 6 dropped out after the fifth treatment session.

The small number of participants and the fact that one of them dropped out during treatment are important limitations of this study. The type of design used is also a methodological limitation because it is not possible to draw firm conclusions from a series of case observations. Despite these limitations, this case series yielded some interesting observations. First, it seems that it is possible to address GAD among adolescents with interventions derived from a specific conceptualization of excessive worry. Considering the three assessment modalities, it seems likely that participants 1, 4, and 7 truly benefited from the treatment program. The worry interventions seemed to be very well understood by these adolescents and were applied through homework to their daily lives. Conversely, the low treat-

TABLE 1
Outcome Measures Scores at Pretest, Posttest and Follow-up Assessments

Participants	Measures	Pretest	Posttest	6-Month Follow-up	12-Month Follow-up
1	GAD	Marked (6/8)	r.s. 3/8	r.s. 4/8	r.s. 5/8
	PSWQ	71	51	53	62
	WAQ	5	2	5	4
	WDQ	118	72	87	74
	BAI	36	14	4	14
	CDI	28	14	16	21
	S-M (%)	71	20	22	26
2	GAD	Marked-severe (7/8)	Moderate (4/8)	Moderate (4/8)	Moderate (4/8)
	PSWQ	73	63	57	46
	WAQ	6	5	6	6
	WDQ	96	78	76	63
	BAI	20	15	25	28
	CDI	25	22	25	14
	S-M (%)	64	29	37	25
3	GAD	Marked (6/8)	Marked (6/8)	—	Marked (6/8)
	PSWQ	71	62	—	—
	WAQ	6	6	—	—
	WDQ	63	86	—	—
	BAI	31	26	—	—
	CDI	30	26	—	—
	S-M (%)	52	47	—	—
4	GAD	Marked (6/8)	r.s. 3/8	r.s. 2/8	r.s. 2/8
	PSWQ	69	59	39	36
	WAQ	3	0	1	0
	WDQ	81	40	30	13
	BAI	27	16	4	0
	CDI	21	11	6	5
	S-M (%)	61	33	13	11
5	GAD	Marked (6/8)	Moderate (4/8)	Moderate-marked (5/8)	r.s. 6/8
	PSWQ	58	54	63	44
	WAQ	5	0	4	6
	WDQ	116	109	122	27
	BAI	34	14	34	22
	CDI	29	20	26	24
	S-M (%)	63	40	44	51
6	GAD	Moderate-marked (5/8)	—	—	—
	PSWQ	70	—	—	—
	WAQ	4	—	—	—
	WDQ	49	—	—	—
	BAI	26	—	—	—
	CDI	21	—	—	—
	S-M (%)	38	—	—	—
7	GAD	Moderate-marked (5/8)	r.s. 3/8	r.s. 2/8	r.s. 2/8
	PSWQ	61	55	41	44
	WAQ	5	3	3	2
	WDQ	95	61	38	53
	BAI	19	15	7	15
	CDI	23	12	8	9
	S-M (%)	28	11	25	7

Note: GAD = presence and severity of generalized anxiety disorder diagnosis; when the diagnosis is absent, the residual symptoms (r.s.) of GAD are rated; PSWQ = Penn State Worry Questionnaire; WAQ = Worry and Anxiety Questionnaire (number of GAD somatic symptoms at a severity of 4/8 or greater); WDQ = Worry Domains Questionnaire; BAI = Beck Anxiety Inventory; CDI = Children's Depression Inventory; S-M = daily self-monitoring of worry.

TABLE 2

Number of Participants in Each Category of Responder Status at Posttest and Follow-up Assessments

No. of Measures	Posttest ^a	6-Month Follow-up ^b	12-Month Follow-up ^b
0–1	1	0	0
2–4	2	2	2
5–7	3	3	3

^a *n* = 6.

^b *n* = 5.

ment response of participants 2, 3, and 5 might have been due to the psychoeducational style of the therapy which supplied a certain amount of new information and required the use of notebooks and homework. For example, participant 2 was going to college but was experiencing serious concentration difficulties during her classes and procrastination problems with her homework. Participant 3 was registered part-time at college, but was not regularly attending his classes, whereas participant 5 had not finished high school and was encountering problems with obtaining his professional diploma. It is possible that difficulties with the school educational system experienced by these older adolescents had a negative impact on therapy with similar structural features. Even though the cognitive-behavioral approach used in this study involved a simple and adapted vocabulary to explain treatment, it may be that the format is less appealing to adolescents who do not enjoy school. The way the worry interventions are presented might be further adapted to more closely fit with the age and maturity of the adolescent undertaking treatment of GAD. One of the challenges of future research is to use a controlled group design while keeping the treatment sufficiently individualized for each adolescent.

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